

Pythagoras and Trigonometry

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Complete Pythagoras' theorem for a right-angled triangle with shorter sides a and b and hypotenuse c .

$$a^2 + \square = \square$$

1 The two shorter sides squared add up to the hypotenuse squared.

2 The hypotenuse is always the longest side, opposite the right angle.

Answer: $a^2 + b^2 = c^2$

2 [2 marks]

A right-angled triangle has shorter sides 6 cm and 8 cm.

Work out the length of the hypotenuse.

1 $6^2 + 8^2 = 36 + 64 = 100$

2 Hypotenuse = $\sqrt{100} = 10$ cm

Answer: 10 cm

3 [2 marks]

A right-angled triangle has shorter sides 9 cm and 12 cm.

Work out the length of the hypotenuse.

1 $9^2 + 12^2 = 81 + 144 = 225$

2 $\sqrt{225} = 15$ cm

Answer: 15 cm

4 [2 marks]

A right-angled triangle has hypotenuse 13 cm and one shorter side 5 cm.

Work out the length of the other shorter side.

1 To find a shorter side you **subtract**: $13^2 - 5^2$

2 $169 - 25 = 144$

3 $\sqrt{144} = 12$ cm

Answer: 12 cm

5

[3 marks]

A right-angled triangle has shorter sides 7 cm and 10 cm.

Work out the hypotenuse, correct to 1 decimal place.

$$1 \quad 7^2 + 10^2 = 49 + 100 = 149$$

$$2 \quad \sqrt{149} = 12.2065\dots$$

3 Rounded to 1 decimal place that is 12.2 cm.

Answer: 12.2 cm (1 d.p.)

6

[3 marks]

A right-angled triangle has hypotenuse 15 cm and one shorter side 9 cm.

(a) Work out the other shorter side.

(b) Work out the area of the triangle.

$$1 \quad (a) \quad 15^2 - 9^2 = 225 - 81 = 144, \text{ so the side is } \sqrt{144} = 12 \text{ cm}$$

2 (b) The two shorter sides are perpendicular, so use them as base and height.

$$3 \quad \text{Area} = \frac{1}{2} \times 9 \times 12 = 54 \text{ cm}^2$$

Answer: (a) 12 cm (b) 54 cm²

7

[3 marks]

In a right-angled triangle the hypotenuse is 10 cm and one angle is 40°.

Work out the side opposite the 40° angle, to 1 decimal place.

1 You know the hypotenuse and want the opposite side, so use sine.

$$2 \quad \sin 40^\circ = \frac{\text{opposite}}{10}$$

$$3 \quad \text{opposite} = 10 \times \sin 40^\circ = 10 \times 0.642787\dots$$

$$4 \quad = 6.4278\dots \text{ cm}$$

Answer: 6.4 cm (1 d.p.)

8

[3 marks]

In a right-angled triangle the hypotenuse is 12 cm and one angle is 25°.

Work out the side adjacent to the 25° angle, to 1 decimal place.

1 Hypotenuse and adjacent means cosine.

$$2 \quad \cos 25^\circ = \frac{\text{adjacent}}{12}$$

$$3 \quad \text{adjacent} = 12 \times \cos 25^\circ = 12 \times 0.906307\dots$$

$$4 \quad = 10.8757\dots \text{ cm}$$

Answer: 10.9 cm (1 d.p.)

9

[4 marks]

In a right-angled triangle, the side opposite angle θ is 5 cm and the hypotenuse is 9 cm.
Work out the size of angle θ , to 1 decimal place.

1 Opposite and hypotenuse means sine.

$$2 \quad \sin \theta = \frac{5}{9} = 0.5555\dots$$

3 To find the angle, use the inverse sine function.

$$4 \quad \theta = \sin^{-1}(0.5555\dots)$$

$$5 \quad = 33.749\dots^\circ$$

Answer: $\theta = 33.7^\circ$ (1 d.p.)

10

[4 marks]

A ladder of length 5 m leans against a vertical wall. The foot of the ladder is 1.4 m from the base of the wall.

(a) How far up the wall does the ladder reach? Give your answer to 1 decimal place.

(b) Work out the angle between the ladder and the ground, to 1 decimal place.

1 (a) The wall, ground and ladder form a right-angled triangle with the ladder as the hypotenuse.

$$2 \quad \text{height}^2 = 5^2 - 1.4^2 = 25 - 1.96 = 23.04$$

$$3 \quad \text{height} = \sqrt{23.04} = 4.8 \text{ m exactly}$$

4 (b) The 1.4 m is adjacent to the angle and the 5 m is the hypotenuse, so use cosine.

$$5 \quad \cos \theta = \frac{1.4}{5} = 0.28$$

$$6 \quad \theta = \cos^{-1}(0.28) = 73.739\dots^\circ$$

Answer: (a) 4.8 m (b) 73.7° (1 d.p.)

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A right-angled triangle has hypotenuse 17 cm and one shorter side 8 cm.
Work out the other shorter side.

$$1 \quad 17^2 - 8^2 = 289 - 64 = 225$$

$$2 \quad \sqrt{225} = 15 \text{ cm}$$

Answer: 15 cm

2

[2 marks]

Work out the distance between the points (1, 2) and (5, 5).

$$1 \quad \text{Horizontal difference} = 5 - 1 = 4 \text{ and vertical difference} = 5 - 2 = 3$$

$$2 \quad \text{Distance}^2 = 4^2 + 3^2 = 25, \text{ so the distance is } 5$$

Answer: 5

3

[3 marks]

In a right-angled triangle the side opposite angle θ is 8 cm and the side adjacent is 6 cm.
Work out θ , to 1 decimal place.

1 Opposite and adjacent means tangent.

2 $\tan \theta = \frac{8}{6} = 1.3333\dots$

3 $\theta = \tan^{-1}(1.3333\dots)$

4 $= 53.130\dots^\circ$

Answer: $\theta = 53.1^\circ$ (1 d.p.)

4

[3 marks]

Write down the exact value of

(a) $\sin 30^\circ$ (b) $\cos 60^\circ$ (c) $\tan 45^\circ$

1 These come from the standard triangles you are expected to know.

2 (a) $\sin 30^\circ = \frac{1}{2}$

3 (b) $\cos 60^\circ = \frac{1}{2}$

4 (c) $\tan 45^\circ = 1$

Answer: (a) $\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 1

5

[4 marks]

A cuboid measures 6 cm by 8 cm by 10 cm.

Work out the length of the longest straight line that can be drawn inside it, to 1 decimal place.

1 The longest line is the space diagonal, from one corner to the opposite corner.

2 First find the diagonal across the 6 by 8 face: $6^2 + 8^2 = 100$, so it is 10 cm.

3 Now use that as one side of a right-angled triangle with the 10 cm edge.

4 $\text{diagonal}^2 = 10^2 + 10^2 = 200$

5 $\sqrt{200} = 14.142\dots$ cm

6 (A shortcut is $\sqrt{6^2 + 8^2 + 10^2} = \sqrt{200}$.)

Answer: 14.1 cm (1 d.p.)

6

[4 marks]

In triangle ABC, angle A = 50° , angle B = 35° and side a (opposite A) = 12 cm.

Work out side b, to 3 significant figures.

1 Two angles and the side opposite one of them means the sine rule.

$$2 \quad \frac{a}{\sin A} = \frac{b}{\sin B}, \text{ so } b = \frac{a \times \sin B}{\sin A}$$

3 $\sin 35^\circ = 0.573576$ and $\sin 50^\circ = 0.766044$

$$4 \quad b = \frac{12 \times 0.573576}{0.766044} = \frac{6.882917}{0.766044}$$

$$5 = 8.9850... \text{ cm}$$

Answer: $b = 8.99 \text{ cm}$ (3 s.f.)

7

[4 marks]

A triangle has sides 5 cm, 7 cm and 9 cm.

Work out the size of the largest angle, to 1 decimal place.

1 The largest angle is opposite the longest side, so it is opposite the 9 cm side.

$$2 \quad \text{Use the cosine rule rearranged: } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$3 \quad \cos C = \frac{5^2 + 7^2 - 9^2}{2 \times 5 \times 7} = \frac{25 + 49 - 81}{70}$$

$$4 \quad = \frac{-7}{70} = -0.1$$

5 A negative cosine means the angle is obtuse.

$$6 \quad C = \cos^{-1}(-0.1) = 95.739...^\circ$$

Answer: 95.7° (1 d.p.)

8

[4 marks]

A triangle has two sides of 8 cm and 11 cm with an angle of 47° between them.

Work out the area of the triangle, to 3 significant figures.

1 Use Area = $\frac{1}{2}ab \sin C$, where C is the angle between the two sides.

$$2 \quad \text{Area} = \frac{1}{2} \times 8 \times 11 \times \sin 47^\circ$$

$$3 \quad \frac{1}{2} \times 8 \times 11 = 44$$

$$4 \quad \sin 47^\circ = 0.731354...$$

$$5 \quad 44 \times 0.731354 = 32.179... \text{ cm}^2$$

Answer: 32.2 cm^2 (3 s.f.)

9

[4 marks]

A ship sails 40 km on a bearing of 070° , then 30 km on a bearing of 160° .

(a) Work out how far the ship now is from its starting point.

(b) Work out the bearing of the ship from its starting point, to the nearest degree.

1 (a) The bearings differ by $160 - 70 = 90^\circ$, so the two legs are at right angles.

2 $\text{distance}^2 = 40^2 + 30^2 = 1600 + 900 = 2500$

3 $\sqrt{2500} = 50 \text{ km}$

4 (b) In that right-angled triangle, the angle at the start between the first leg and the direct line is $\tan^{-1}\left(\frac{30}{40}\right) = 36.87^\circ$

5 The direct line is that much further clockwise than the 070° bearing.

6 Bearing = $070 + 36.87 = 106.87^\circ$, so 107° to the nearest degree.

Answer: (a) 50 km (b) 107°

10

[5 marks]

A pyramid has a square base ABCD of side 8 cm. The apex E is directly above the centre of the base, and $EA = 10 \text{ cm}$.

(a) Work out the length of the base diagonal AC, to 2 decimal places.

(b) Work out the vertical height of the pyramid, to 2 decimal places.

(c) Work out the angle EA makes with the base, to 1 decimal place.

1 (a) AC is the diagonal of an 8 cm square: $AC^2 = 8^2 + 8^2 = 128$

2 $AC = \sqrt{128} = 11.3137\dots \text{ cm}$, so 11.31 cm

3 (b) The apex is above the centre, so the horizontal distance from A to that point is half of AC = 5.6568... cm

4 $\text{height}^2 = 10^2 - 5.6568^2 = 100 - 32 = 68$

5 $\text{height} = \sqrt{68} = 8.2462\dots \text{ cm}$, so 8.25 cm

6 (c) In that right-angled triangle the adjacent side is 5.6568 and the hypotenuse is 10.

7 $\cos \theta = \frac{5.6568}{10}$, so $\theta = \cos^{-1}(0.56568) = 55.550\dots^\circ$

Answer: (a) 11.31 cm (b) 8.25 cm (c) 55.6°